

Sai Mun Lee

SOFTWARE ENGINEER, PRODUCT | FULL STACK

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Summary

Senior Software Engineer with 8+ years shipping large-scale, full-stack products end to end — from API design and frontend delivery to production reliability. Leads major zero-to-one initiatives and cross-functional teams (product, design, infra), owning components with expert end-to-end understanding. Pairs strong product-engineering fundamentals (code review, testing, monitoring, safe rollout) with deep experience building and serving ML-powered features at enterprise scale (8B+ recommendations served).

Skills

Languages	Python, TypeScript, JavaScript, Go, Bash
Frontend	React, Next.js, Redux, Jest
Backend	FastAPI, Flask, Django, Express.js, PostgreSQL, HANA DB, ElasticSearch
Infrastructure	Kubernetes, Docker, Cloud Foundry, Jenkins, Terraform
Practices	System Design, API Design, Code Review, TDD, Monitoring & Rollout, Agile
ML & AI	Recommender Systems, Foundation Model Serving, PyTorch, LLM & RAG (prototyping)
Spoken	English, Chinese

Experience

SAP Asia

Singapore

SENIOR SOFTWARE ENGINEER | CO-PRODUCT OWNER

Dec 2020 - Present

- Led a 15-person engineering team across frontend and backend to deliver Personalized Recommendations end to end, a product feature that generated 8B+ job recommendations in 2025 for enterprise customers on SAP SuccessFactors.
- Set technical direction and goals for the team, established ownership of core components, and drove major initiatives from zero to one under high ambiguity.
- Architected from scratch SAP's Tabular AI platform — a multi-tenant serving layer targeting 10+ enterprise tenants at 100K users each, owning API contracts, tenant isolation, and system architecture, deployed on Kubernetes.
- Collaborated with cross-functional partners (product, design, infrastructure) to translate requirements into scalable application experiences, running weekly design reviews and stakeholder sessions.
- Built developer-facing FastAPI and Flask APIs with comprehensive documentation, enabling seamless integration across SAP's ecosystem, and delivered React/Redux frontends for internal-facing tools.
- Drove quality and reliability through code review, unit/integration/e2e testing, and CI/CD (Jenkins); instrumented the platform with Dynatrace APM and Kibana correlation-ID tracing, using the telemetry to diagnose production issues and reduce mean-time-to-resolution.
- Analyzed and optimized the model inference serving path to resolve latency and throughput bottlenecks under concurrent multi-tenant load, and mentored engineers through pair programming and code reviews.

TAU Express

Singapore

LEAD SOFTWARE ENGINEER

Apr. 2020 - Dec. 2020

- Led development of a knowledge management system enabling analysts to retrieve and correlate insights from cybersecurity documents using graph-based relationships.
- Owned end-to-end delivery: gathered requirements directly from users, designed system architecture, and facilitated communication between users and the development team.
- Tech: React, Django, Docker, Neo4J, ElasticSearch, Cytoscape.js

SPIRIT Research Center (NTU)

Singapore

RESEARCH ENGINEER

Feb. 2018 - Mar. 2020

- Developed the Intelligent Case Retrieval System (ICRS) for Singapore's smart nation initiative — a full-stack application (Django, Vue.js) for efficient retrieval of legal documents.
- Built web crawlers (Selenium, BeautifulSoup) to index Chinese news articles and social media content for document retrieval pipelines.

Certificates

Dec 2021 **AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)

Feb 2022 **GCP Associate Cloud Engineer**, Google Cloud Platform (GCP)

May 2023 **Hashicorp Certified: Terraform Associate (003)**, Hashicorp

AWS Certified Solutions Architect - Professional, Amazon Web Services (AWS)

In Progress

Nanyang Technological University (NTU)

B.S. IN COMPUTER SCIENCE (HONOURS) — MACHINE LEARNING & DATA SCIENCE SPECIALISATION

Singapore

2013 - 2017